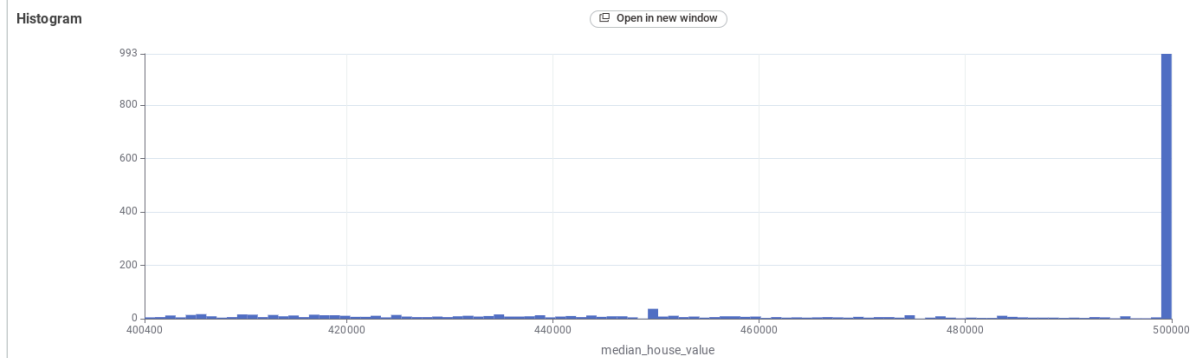


Week 5 — Fix the Model & Optimization

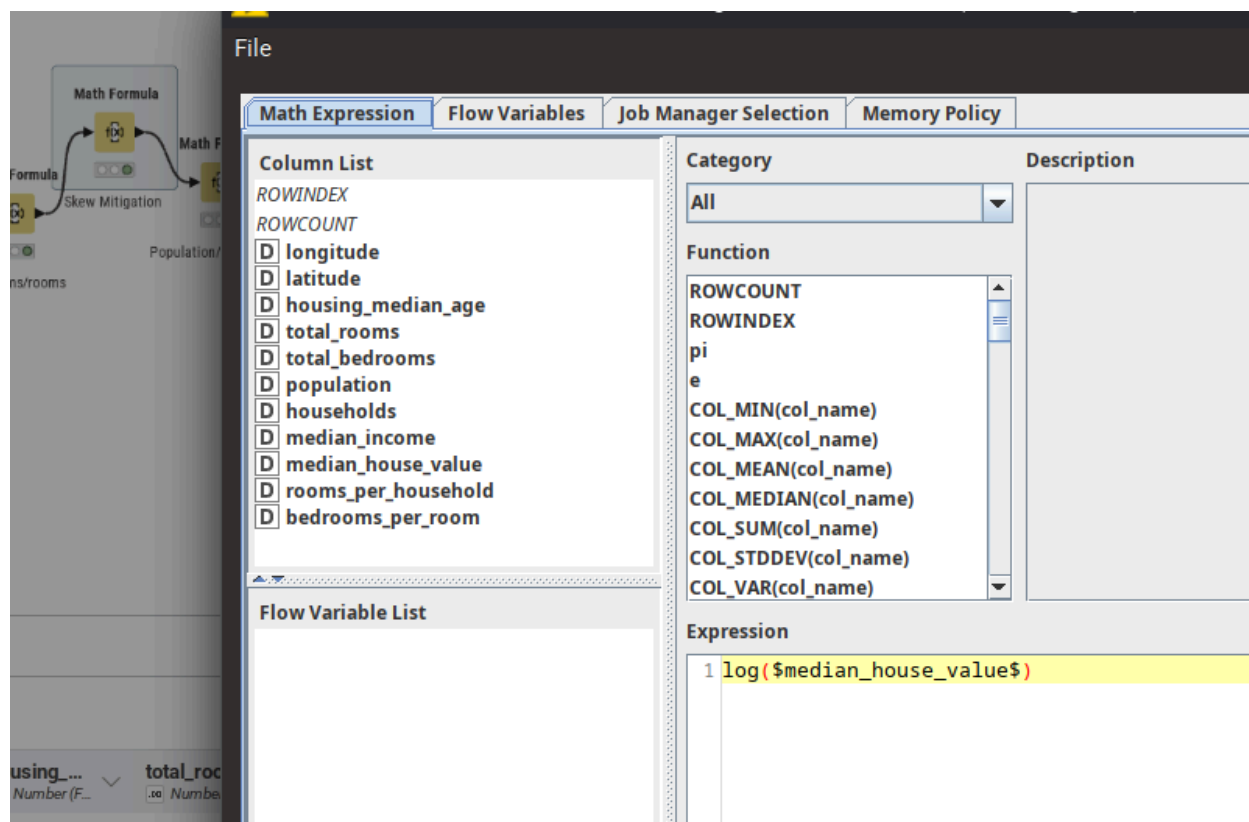
Deliverable 5: Corrected Pipeline. Tasks:

- **Address overprediction above \$400k:** Nodes: Histogram, Row Filter, Math Formula, Numeric Scorer, Tile View, Annotation.

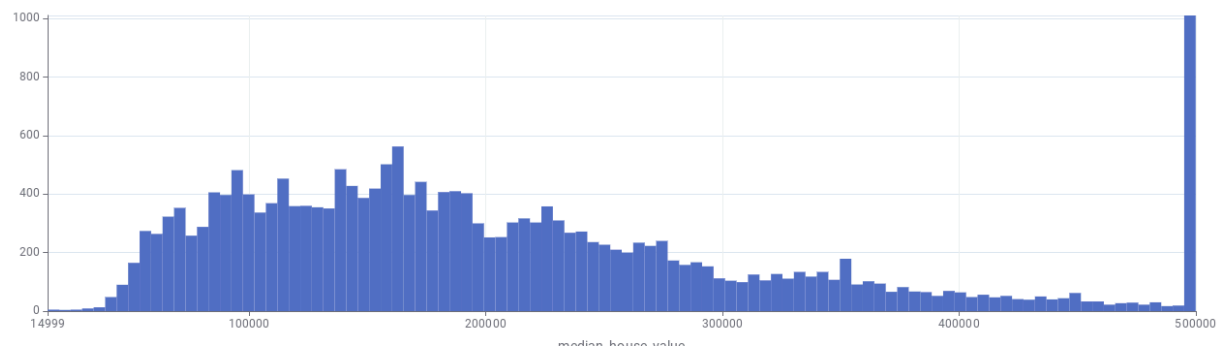


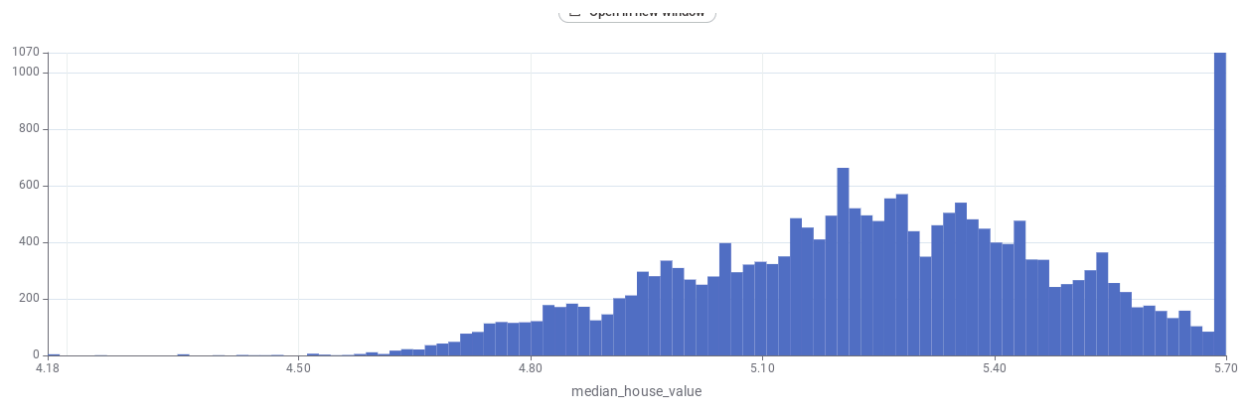
As shown in this histogram, this is a view of all houses above 400k. Due to the cutoff at \$499,000 there will be a severe overprediction of values in this range because of the large spike seen at the 500,000 mark. Using a row filter we can filter out house values that are greater than \$499,000, alleviating the problem by artificially filtering the data out. This will however not represent this data, but it is a problem with the original data set.

- **Skew Mitigation:** Use **Math Formula** (Log transforms) to address skewed histograms found in Week 2.

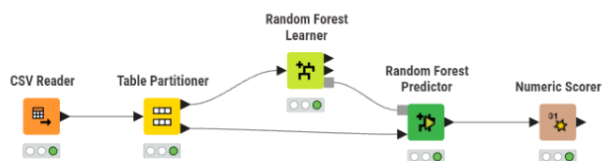
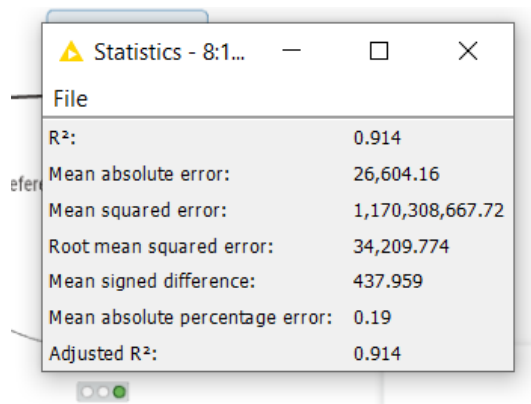


For mitigating the skew of the median house value, we can apply a log transform. This changes the histograms from a skewed diagram to a more uniform (however shifted upward) distribution.





- **Error Reduction:** Implement a **Random Forest Learner**. Your goal is to reduce **MAE below \$35,000**. Nodes: Numeric Scorer, Random Forest Learner, Annotation.



To reduce the MAE below \$35,000, I followed the node instructions to eventually get my reduced value of \$26,604. When I tried using other nodes as my reference column in the scorer node, I was unable to get MAE below the required amount and therefore had to use median_house_value as my target.

This model was able to better capture non-linear relationships in the housing data compared to simpler models. Additionally, preprocessing steps such as skew mitigation helped normalize the data and reduce the influence of extreme values.

During evaluation, the correct configuration of the Numeric Scorer was ensured by comparing the true median_house_value column with the predicted output column generated by the Random Forest Predictor. This allowed for an accurate calculation of MAE and demonstrated improved model performance below the target threshold of \$35,000.

- **Classification Task:** Use **Logistic Regression** to predict if a house is "NEAR OCEAN." Compare results with Random Forest.
Nodes: Rule Engine, Column Filter, One to Many, Logistic Regression Learner, Scorer, Random Forest Learner, Tile View, Annotation.

► 1: Input data and view selection 

Rows: 4 | Columns: 5

Table  Statistics 

☐	#	RowID	No Number (Integer)	Yes Number (Integer)	0 Number (Integer)	1 Number (Integer)	Selected (Tile View) Boolean
☐	1	No	0	0	5420	773	false
☐	2	Yes	0	0	0	0	false
☐	3	0	0	0	0	0	false
☐	4	1	0	0	0	0	false

(above) logistical regression tile view confusion matrix

► 1: Input data and view selection 

Rows: 4 | Columns: 5

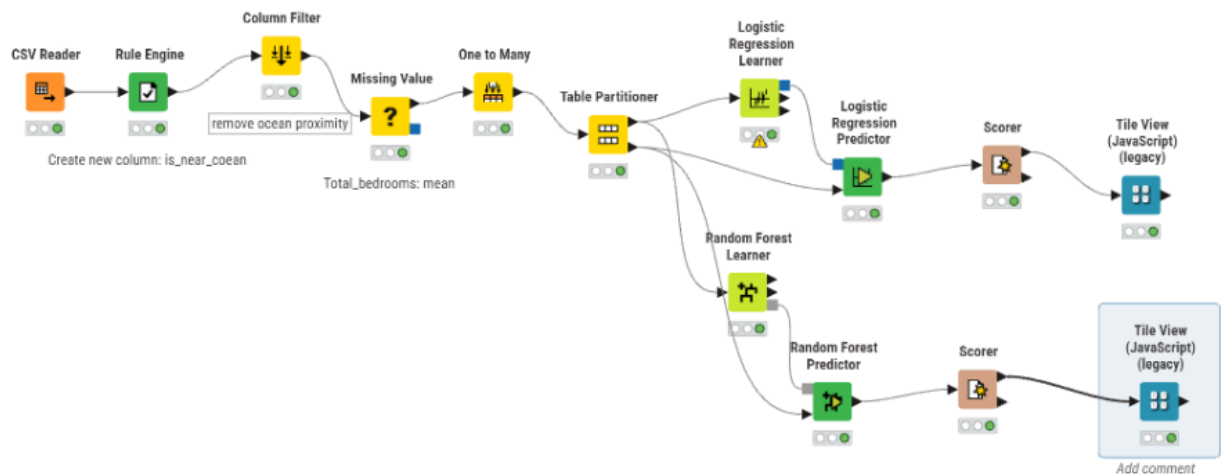
Table  Statistics 

☐	#	RowID	No Number (Integer)	Yes Number (Integer)	0 Number (Integer)	1 Number (Integer)	Selected (Tile View) Boolean
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☐	2	Yes	0	0	0	773	false
☐	3	0	0	0	0	0	false
☐	4	1	0	0	0	0	false

(above) random forest tree tile view confusion matrix

Here I used logistical regression and a random forest tree to predict if a house was near an ocean. Quickly, you can see that logistic regression struggled with this task and was not able to identify the correct instance of "near ocean" classification and mainly predicts the majority of the houses to be "no" (no/1 = 773) and (no/0 = 5420). This gave an obvious poor performance with bad precision since it did not have any true positive predictions (we obviously know there are houses that are near the ocean, so there has to be some kind of "yes"). When looking at the Random Forest model, it performed much better and correctly classified our numbers with no false positives or false negatives. The math seemed to line up on both as they had the same number, just in different categories (random forest has the numbers in the correct categories). This shows us that the random forest was able to capture the complex and non-linear relationship in the data that the logistic regression could not. Overall, it seems like the

random forest is the more accurate model for this classification task as it had correct accuracy and handled the classes effectively.



Team Weekly Sync:

Project Checklist

Group Name/Number: Group 5 | **Date:** 04/19/26

Use this checklist at the start of every meeting to ensure you are meeting the "Green-to-Gold" requirements and project milestones.

1. Infrastructure & Accountability (Start-of-Meeting)

- [x] **GitHub Sync:** Did everyone pull the latest changes from `main`?
- [x] **Contribution Log:** Has the `README.md` been updated to reflect progress made this week?
- [x] **Blocking Issues:** Is anyone currently "blocked" by a KNIME node error or a data issue? *(If yes, resolve this first!)*

2. Current Sprint Focus (Check the Week)

- **Week 5 (Fairness):** Have you compared at least two corrective strategies (e.g., threshold adjustment vs. oversampling)?

yes

3. Technical Quality Assurance (The "Green-to-Gold" Audit)

- [x] **Automation:** Does the workflow run from **File Reader** to **Output** without changing settings?
- [x] **Annotation:** Are your primary logic steps (e.g., normalisation, optimization) explained in a **Metanode Annotation**?
- [x] **Evidence:** Is every chart/table in your report directly linked to a specific KNIME node output?

4. Action Items (End-of-Meeting)

Who is doing what before next week?

Task	Assignee	Deadline
Slideshow and data	Brian Mellinger	4/26
Slideshow and data	Rachel Roberts	4/26
Slideshow and data	Isaiah Deppong	4/26